

AFSAL A AZEEZ

Full-Stack & AI Engineer • GenAI • IIT Bombay M.Tech

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Abu Dhabi, UAE • **Available Immediately**

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Professional Summary

Full-Stack & AI Engineer with a profound technical background specializing in designing, building, and scaling robust distributed microservices and production-grade AI/LLM applications. Proven track record at a Silicon Valley venture-backed startup driving technical strategy, optimizing backend APIs, and engineering multi-LLM orchestration platforms. Recently architected and shipped Markaba AI, an AI-powered automotive diagnostics and marketplace platform with a natural-language booking assistant and a golden-set LLM eval harness, solo from first commit to live production. Expert in cloud-native infrastructures (AWS/Kubernetes) and high-performance system design. Graduated from IIT Bombay (GATE AIR-241); fully attested academic credentials for immediate UAE residency compliance.

Technical Skills

- **Architecture & Design:** Microservices, Distributed Systems, RESTful/GraphQL APIs, Serverless, Event-Driven Architecture, System Design.
- **Frontend Engineering:** React.js, Next.js, Electron.js, Redux Toolkit, Zustand, TailwindCSS, SCSS, HTML5/CSS3.
- **Backend Engineering:** Node.js (TypeScript), Python (FastAPI/Django), Go, Express, Fastify, OAuth, JWT.
- **AI & LLM:** GPT-4, Gemini, Claude, Llama 3, LangChain, LlamaIndex, OpenAI API, RAG Pipelines, Vector Databases, Tool-Calling Agents, Model Context Protocol (MCP), LLM Evaluation, Prompt Caching, Multi-LLM Orchestration, AI Agent Orchestration.
- **DevOps & Cloud:** AWS (EC2, S3, Lambda, EKS), Docker, Kubernetes, OpenShift, Terraform, Helm, CloudFormation, GitHub Actions, Jenkins.
- **Databases & Caching:** PostgreSQL, MySQL, MongoDB, Redis, SQLite, Elasticsearch.
- **Testing & Security:** Playwright, OWASP ZAP, Postman, Artillery, Selenium, JUnit, pytest, Jest.
- **Version Control & Platforms:** GitHub, GitLab, Bitbucket, Azure Repos, Microsoft Teams (Bot Framework).

Professional Experience

Founder & Lead Engineer

Jul 2026 – Present

Full-Stack & AI Engineer

Markaba AI (markabaai.com) — AI Automotive Diagnostics Platform

- **AI Diagnosis Engine:** Architected a multi-layer automotive intelligence engine (Diagnose → Match → Rank → Quote) converting natural-language vehicle symptoms into ranked repair plans with matched parts and bookable workshop quotations; runs on Claude in production (eval-validated Gemini adapter, switchable at runtime) behind a circuit breaker, Claude prompt caching (42% lower cost on a live eval run), and a daily AI-spend cap.
- **AI Booking Assistant:** Customers who name a garage can book it in plain language: the LLM extracts a structured request, and the backend resolves provider and slot against live data, prices it server-side, and asks rather than guesses when anything is ambiguous; the customer confirms via the existing booking endpoint, and it offers to book only when asked (eval-checked).
- **Tool-Calling Agent & Evals:** Built a 33-case golden-set eval harness (diagnosis, photo ID, voice; incl. Arabic) to compare Claude vs Gemini, and A/B-tested a bounded tool-calling agent (5 tools; turn, tool-call, and timeout caps) against the production pipeline — no quality gain on 4 ambiguous cases at 2× latency and 3.6× cost, so it stays admin-only.
- **MCP Connectors (live in production):** Two Model Context Protocol servers letting Claude read Markaba AI — car-owner (7 read-only tools) and business (5) — OAuth resource servers with Clerk as the authorization server, prompt-injection mitigations for third-party text, per-tool output allowlists, and fail-closed rate limits; on for every account since Sep 2026.
- **Semantic Parts Matching:** Built a RAG-based parts-matching pipeline using LlamaIndex and Ollama embeddings over a PostgreSQL + pgvector catalog, with keyword-matching fallback for reliability.
- **Full-Stack Marketplace:** Shipped a bilingual (English/Arabic, RTL) React + Vite + TypeScript + Tailwind frontend and a FastAPI + SQLAlchemy + Alembic backend covering a parts marketplace, “Mod Discovery” upgrades, service booking, a “My Garage” vehicle-profile system, and provider and supplier portals with wholesale procurement; wrapped with Capacitor for Android (submitted to Google Play closed testing) and iOS (simulator).
- **Parts Network (built; pilot pending):** B2B layer where workshops trade parts and quote repairs — part requests ranked by location and price, repair plans quoted by up to three workshops, and accepted jobs that book a

slot and raise one purchase order per seller. Forward-only Postgres state machines with arq workers, not LangGraph (the edges are business rules), plus a bilingual part-identity resolver where the model only suggests (61-case eval).

- **Payments & Notifications:** Built a Tap Payments integration, Clerk/OIDC authentication, and SendGrid/Twilio (email, SMS, WhatsApp) workflows, including a WhatsApp Accept/Decline flow for partner workshops.
- **Solo Production Delivery:** Owned architecture, backend, frontend, and DevOps end-to-end (Docker, Caddy on a VPS, GitHub Actions CI, and a one-command, health-gated deploy script), from zero to live production within two weeks of the first commit; since load-tested locally with k6 and moved to two backend replicas with rolling deploys, Postgres advisory locks, deploy rollback, and off-box backups.
- **Tech Stack:** Python, FastAPI, React, Vite, TypeScript, Tailwind CSS, Capacitor, PostgreSQL, pgvector, Redis, arq, LlamaIndex, Docker, GitHub Actions, k6, Anthropic Claude API, Google Gemini API.

Member of Technical Staff

Sep 2020 – May 2026

Full-Stack & AI Engineer Zettabytes Inc. (Roost.ai) — Mayfield-funded DevOps & AI Startup · Remote / US-Based Market

Phase B — RoostGPT: AI Test Generation Platform

2023 – 2026

- **RoostGPT — AI Testing Platform:** Built and shipped **RoostGPT**, an AI testing copilot leveraging LLMs (GPT-4, Gemini, Claude, Llama 3) to auto-generate unit, API, UI, and integration test cases from source code, Swagger/OpenAPI specs, and Jira user stories.
- **Microsoft Teams Notification Bot:** Took **end-to-end ownership** of building a Microsoft Teams notification bot for RoostGPT, delivering real-time test generation status updates, results summaries, and CI/CD pipeline notifications directly within Teams channels.
- **Git Platform Integrations:** Built integrations within RoostGPT connecting to GitHub, GitLab, Bitbucket, and Azure Repos for automated repository access, test generation triggers, and direct test commit workflows. Implemented GitHub pull request webhook hooks to automatically retrigger test generation on new code changes.
- **Multi-LLM Provider UI:** Built the frontend dashboard for configuring and switching between LLM providers — supporting OpenAI, Google Gemini, Anthropic Claude, and Meta Llama 3 as interchangeable backends, enabling customers to select their preferred AI model for test generation.
- **Java Unit Test Generation — End-to-End Ownership:** Owned the complete Java unit test generation pipeline — from repository ingestion through to final generated test files. Engineered LLM context by working directly on the **Spring Boot Java AST parsing service** to extract and structure class graphs, bean dependency trees, and method signatures from multi-module Maven/Gradle repositories. Diagnosed and resolved prompt engineering issues causing incorrect imports and missing mocks, achieving a **93% compile success rate** across generated test suites.
- **Test Scenario Analysis & Report UI:** Built the end-to-end test scenario analysis feature (frontend, API, and backend) — listing all generated tests with covered and uncovered scenarios, allowing users to select specific scenarios and trigger targeted test generation. Displayed generation summaries, compile check results, and parsed coverage report output, giving developers a full view of test quality before review and commit. Achieved **4x improvement in test coverage** and **85% reduction in manual testing time**.
- **Prompt Management System:** Designed and shipped a **nested-map data structure** for managing LLM prompts across multiple dimensions — model (GPT-4, Gemini, Claude, Llama 3), test type (unit, API spec, integration), and client. Implemented a deterministic lookup algorithm enabling the system to fetch the exact prompt for any combination at runtime, and enforced structured folder conventions for prompt files with **AWS S3-backed storage** for versioning and distribution.
- **Java AST Schema Redesign:** Flattened nested class structures into first-class entities, cutting per-class file size from **316KB to 17KB** (95% reduction) and improving LLM context efficiency; validated on ask-java and adk-java.
- **FQN Mapping & Method Overload Resolution:** Implemented Fully Qualified Name mapping for internal method calls, enabling accurate method overload identification via argument lists.
- **File-to-Class Entity Migration:** Led migration from file-based to class-based entity handling across the test generation pipeline, enabling AI agents to correctly receive structured parameters for Java unit test generation.
- **AST Output Optimisation:** Removed unnecessary body fields from method call extraction, reducing codeparse output from **15.9MB to significantly smaller payloads** — cutting LLM token costs and improving throughput.
- **Test Generation UI:** Built the full frontend for test generation workflows — repository analysis views, trigger dashboards, and review/refinement interfaces for inspecting, editing, and approving AI-generated tests.
- **User Access Management UI:** Built team-level permissions for test generation across repositories and projects.
- **AI Response Caching:** Designed an SQLite-based caching layer for LLM responses to eliminate redundant API calls, reducing cost and latency for repeated prompts.
- **Playwright E2E Testing:** Wrote and reviewed Playwright end-to-end test suites and built an API test analysis page for scenario-specific execution tracking.
- **Technical Mentorship:** Mentored junior engineers and interns; conducted code reviews and tracked delivery via

JIRA.

- **Tech Stack:** React.js, Next.js, TypeScript, Node.js, Spring Boot (Java), SQLite, REST APIs, LLM APIs (OpenAI, Gemini, Claude, Llama 3), Docker, GitHub Actions.

Phase A — Roost Desktop, Serverless & EaaS Platform

2020 – 2022

- **Roost Desktop Application:** Built and maintained a **cross-platform Electron.js + React.js** desktop application for DevOps collaboration targeting macOS, Windows, and Linux.
- **Kubernetes Visualisation:** Developed frontend UI for visualising local Kubernetes deployment environments — pod network topology, sidecar/service-mesh connections, and real-time cluster state monitoring.
- **Serverless Migration:** Transitioned the Roost Desktop product to a **serverless cloud architecture**, building the Roost Web UI as a browser-based interface extending the desktop app's capabilities.
- **Collaboration Features:** Implemented a collaboration layer enabling developers to share environment snapshots and deployment states with QA, product, and external stakeholders for review.
- **EaaS Feature Development:** Implemented multi-repo commit (Save All), repo-level config saving, environment variable injection, Helm chart filtering, CloudFormation (S3/JSON) integration, and Terraform UI with full API integration.
- **GitHub API Integration:** Built integrations with GitHub API for repository file discovery, deployment config parsing, and Kubernetes/Helm YAML filtering for the EaaS config workflow.
- **Slack & JIRA Config UI:** Added Slack token/channel suggestion dropdowns and JIRA-referenced feature tracking fields across the EaaS configuration UI.
- **Enterprise Deployments:** Supported on-premise and OpenShift deployments for enterprise clients across the US market, collaborating directly with their engineering and DevOps teams on cluster integration and platform rollout.
- **Tech Stack:** React.js, Node.js, Python, Go, REST/GraphQL APIs, Kubernetes, Docker, Terraform, AWS (EKS, EC2, S3, Lambda), Electron.js.

Education

Master of Technology (M.Tech.) in Computer Science & Engineering

Graduated 2020

Indian Institute of Technology (IIT), Bombay

Mumbai, India

- GATE AIR-241 — Top 0.22% out of 107,893 candidates. Degree and academic credentials fully attested for UAE international visa and residency compliance.

Bachelor of Technology (B.Tech.) in Computer Science & Engineering

Graduated 2017

Mar Athanasius College of Engineering (MACE)

Kerala, India

- Completed B.Tech in CSE with high academic standing.

Technical, AI & Academic Projects

DTwin — Building Digital Twin with an AI Copilot

Personal Project

- **Digital Twin Platform:** Commercial building on simulated telemetry — PostGIS + TimescaleDB, Node.js WebSocket ingest, FastAPI energy simulator, react-three-fiber 3D dashboard; multi-tenant via row-level security.
- **Human-Approved AI Copilot:** LangGraph.js + Claude agent that cannot change a setpoint without human approval — its command node is reachable only via an `interrupt()`, asserted by a graph test; 293 end-to-end checks in CI.

VisionLog-MLX

Personal / Open-Source

- A MLX-VLM powered vision logging and analysis system utilizing the Qwen2-VL-2B-Instruct model to analyze camera feeds and capture events.

GIS-Integrated Water Network Database System

M.Tech Project – IIT Bombay

- Designed and developed a GIS-integrated water network database application in QGIS using Python and SQL. Programmed custom Python plugins to automate node elevation extraction from contour files and web APIs, and built efficient data extraction workflows for EPANET and JalTantra.

Teaching Assistantships

Graduate Teaching Assistant

Jul 2018 – Nov 2019

Indian Institute of Technology (IIT), Bombay

Mumbai, India

- **Software Lab (CS699):** Guided M.Tech and Ph.D. students in programming assignments, managed lab sessions, and performed continuous evaluation and grading of lab work under Prof. Kavi Arya.
- **Computer Programming (CS101):** Led lab sections for 400+ undergraduate students in a team of 45+ TAs under Prof. Om P. Damani and Prof. Ganesh Ramakrishnan; mentored 14 students and evaluated exams.